MEDISENSE

Low-Cost Vital Signs Monitoring for Low-Resource Healthcare Settings

Master's Thesis Proposal

Ramya Subramanian Porselva Bharathi

M.Sc. Web and Data Science

University of Koblenz, Germany

May 2026

1. EXECUTIVE SUMMARY

Approximately 3 billion people globally lack access to reliable healthcare diagnostics. In rural clinics across India, Sub-Saharan Africa, and Southeast Asia, critical vital signs (heart rate, respiratory rate, blood oxygen saturation, body temperature) go unmeasured or are measured inaccurately because commercial medical monitors cost €500-5,000, far beyond available budgets.

The consequence is severe: children with pneumonia die because elevated respiratory rates go undetected. Elderly patients with sepsis deteriorate fatally because abnormal heart rates are not monitored. Conservative estimates suggest that if early detection through vital signs monitoring prevented just 1% of these deaths, **150,000+ lives could be saved annually**.

MediSense addresses this gap as a social entrepreneurship initiative, enabling low-cost sensors (€50-80) to achieve medical-grade accuracy through intelligent software. The core innovation applies transfer learning from industrial quality control to healthcare sensor validation. This approach leverages my two years of experience in ETL pipeline development and data quality validation for enterprise-scale financial systems at CIBC (Canadian Imperial Bank of Commerce), one of Canada's Big Five banks, where I built systems to detect data inconsistencies, validate transformation logic, and ensure 0.98/10 bug scores across billions of records. These same principles of rigorous data validation, quality control, and anomaly detection in noisy data environments transfer directly to making low-cost medical sensors clinically reliable.

This thesis will demonstrate proof-of-concept through dataset validation, achieving ≥90% accuracy in vital signs monitoring and establishing the foundation for clinical deployment that could save 480+ lives annually by 2031.

2. THE PROBLEM: THE GLOBAL HEALTHCARE GAP

2.1 Scale and Impact

Three billion people (40% of the global population) lack access to basic healthcare diagnostics. These populations are concentrated in rural areas where clinics face severe resource constraints: intermittent electricity, no stable internet connectivity, limited medical equipment budgets, and shortages of trained specialists.

Condition	Annual Deaths (WHO/Lancet)
Pneumonia (children)	~800,000
Sepsis (all ages)	~11 million
Respiratory diseases	~4 million

Early detection of abnormal vital signs could prevent a significant proportion of these deaths. Basic monitoring, if available and accurate, enables timely intervention before conditions become critical.

2.2 Existing Solutions and Their Limitations

Commercial medical monitors (€500-5,000) are unaffordable for low-resource clinics, require stable electricity and technical maintenance unavailable locally, and are designed for hospital environments rather than rural settings.

Low-cost alternatives (€50-100) exist but fail clinically: cheap sensors produce noisy, inaccurate data; without intelligent signal processing, they generate high false alarm rates; consequently, healthcare workers do not trust them. Several research efforts have explored camera-based vital signs monitoring (rPPG) and wearable sensors, achieving 95%+ accuracy in controlled laboratory settings. However, these systems degrade to 70-80% accuracy in real-world conditions due to poor lighting, patient movement, and diverse skin tones. Existing approaches also lack robust methods for distinguishing equipment malfunctions from genuine patient emergencies, leading to alert fatigue.

The critical gap is not hardware availability. It is **the absence of software intelligence to make affordable sensors clinically reliable in challenging real-world conditions**. No prior work has applied transfer learning from industrial quality control (where mature sensor validation techniques and abundant datasets exist) to medical vital signs monitoring (where data is scarce and expensive). MediSense fills this specific gap by demonstrating that industrial QC expertise is directly transferable to healthcare.

3. THE SOLUTION: MEDISENSE

3.1 System Overview

MediSense is a low-cost (€50-80), high-accuracy vital signs monitoring system specifically designed for resource-limited clinical settings. It continuously monitors heart rate, respiratory rate, blood oxygen saturation, and body temperature in real-time with medical-grade accuracy.

Layer	Components
Hardware	Camera (rPPG for HR/RR), pulse oximeter (SpO2/HR), IR temperature sensor, Raspberry Pi microcontroller
Software	Signal processing, transfer learning anomaly detection, clinical alert generation (THESIS FOCUS)
Interface	Real-time vital signs display, color-coded alerts (green/yellow/red), trend visualization, simple patient management

3.2 Core Innovation: Transfer Learning from Industrial QC to Healthcare

For two years at CIBC (Canadian Imperial Bank of Commerce), one of Canada's Big Five banks, I worked as an ETL Developer building enterprise-scale data pipelines for the Wholesale Credit Data Warehouse, a system handling billions of customer and facility records across 300+ tables. My role required rigorous data quality validation: writing custom SQL test queries to verify every transformation before and after job execution, detecting data inconsistencies across complex datasets, resolving mapping and logic errors, and maintaining a bug score of 0.98 out of 10. I built mechanisms to identify when data pipelines produced anomalous outputs, investigated root causes when validation checks failed, and ensured data integrity in noisy, mission-critical environments.

My key insight: **the same principles of data quality control, validation logic, and anomaly detection in enterprise data pipelines apply directly to medical sensor validation**. Both involve ensuring data reliability despite noise, detecting when systems produce incorrect outputs, validating transformations against expected ranges, and maintaining quality in production environments. Industrial quality control and medical vital signs monitoring share fundamental challenges: time-series data with normal versus anomalous patterns, noisy measurements requiring filtering, sensor drift and calibration issues, and the need for reliability in challenging conditions.

MediSense applies transfer learning: pre-training anomaly detection models on abundant industrial sensor datasets (NASA bearing degradation, turbofan engines, industrial IoT), then fine-tuning on limited medical vital signs data (PhysioNet, MIMIC-III). This dramatically reduces the medical training data required while achieving higher accuracy than training from scratch, a critical advantage for low-resource healthcare settings where labeled medical data is scarce and expensive to obtain.

3.3 Dual-Level Anomaly Detection

MediSense distinguishes between two types of anomalies, increasing clinical trust and reducing false alarms:

Detection Level	Purpose	Target Accuracy
Level 1: Sensor Malfunction	Detect equipment failure (camera obscured, oximeter detached, sensor drift)	≥95%
Level 2: Patient Anomaly	Detect abnormal vital signs (tachycardia, respiratory depression, hypoxemia)	≥90%

Clinical alerts translate detected anomalies into actionable guidance for healthcare workers with limited medical training. Example: 'URGENT: HR >120 BPM and rising. Possible tachycardia or sepsis. Assess patient immediately.' This bridges technical detection and clinical decision-making.

4. TECHNICAL METHODOLOGY

4.1 Research Question

Can transfer learning from industrial quality control datasets to medical vital signs datasets enable low-cost sensors (<€100) to achieve medical-grade accuracy (≥90% agreement with reference devices) in vital signs monitoring?

4.2 Datasets

Domain	Dataset	Description	Size
Medical	MIMIC-III	ICU patient vital signs with ground truth	53,423 admissions
	PhysioNet BIDMC	PPG and respiratory signals	53 patients
Industrial	NASA Bearing	Vibration sensor degradation	4 bearings
	NASA Turbofan	Engine sensor time-series	100 engines

All medical datasets include ground truth measurements from medical-grade equipment, enabling rigorous validation. All industrial datasets contain time-series sensor data with normal operating conditions plus anomalies, mirroring the medical monitoring challenge.

4.3 Transfer Learning Pipeline

Step 1: Pre-training on Industrial Data. Train anomaly detection models (autoencoder architecture) on industrial sensor datasets. The model learns general sensor validation features: normal versus anomalous patterns, sensor drift over time, noise characteristics, and signal degradation patterns.

Step 2: Fine-tuning on Medical Data. Load pre-trained weights and adapt to medical vital signs data. Early layers (general sensor features) are frozen; later layers are trained on medical data to learn domain-specific patterns. This dramatically reduces required medical training data.

Step 3: Baseline Comparison. Train identical architecture from scratch using only medical data (no industrial pre-training). Compare accuracy, data efficiency, and generalization between transfer learning and baseline approaches.

Signal processing pipeline: Raw sensor data undergoes bandpass filtering (0.5-4 Hz for heart rate, 0.1-0.5 Hz for respiratory rate), baseline wander removal, Kalman smoothing, and motion artifact reduction. Features are extracted in statistical (mean, standard deviation, skewness), frequency (FFT for dominant frequencies), and temporal domains (peak detection, interval variability). These features feed into the anomaly detection model, which outputs both an anomaly score and classification.

4.4 Success Metrics and Validation

Metric	Target
Primary: Patient anomaly detection accuracy	≥90%
Transfer learning improvement over baseline	≥5 percentage points
Data efficiency gain	Reach 90% with ≤50% of baseline data
Sensor malfunction detection	≥95%
Real-time inference latency	<100ms per reading

Evaluation protocol: Strict 60/20/20 train/validation/test split. Test set locked until final evaluation. Five-fold cross-validation ensures generalization. Statistical significance testing (McNemar's test, $p<0.05$) confirms improvements. Detailed error analysis classifies false negatives (missed emergencies) and false positives (false alarms) to identify failure modes and clinical acceptability.

Timeline: 24 weeks total. Weeks 1-4: Foundation (literature, datasets, baseline). Weeks 5-12: Model development (pre-training, fine-tuning, optimization). Weeks 13-20: Validation (test evaluation, error analysis, computational profiling). Weeks 21-24: Documentation (thesis writing, code release).

5. EXPECTED IMPACT AND CONTRIBUTIONS

5.1 Scientific Contributions

Novel research contribution: First demonstration that transfer learning from industrial quality control to medical vital signs monitoring is viable. This opens a new research direction where abundant industrial datasets can bootstrap medical AI systems that typically suffer from data scarcity.

Data efficiency quantification: Expected result showing transfer learning achieves 90% accuracy with 50-80% less medical data than training from scratch. Critical for low-resource settings where collecting labeled medical data is expensive and difficult. Open-source reference implementation enables researchers and NGOs worldwide to build upon this foundation.

5.2 Real-World Impact Pathway

This thesis establishes proof-of-concept through dataset validation, demonstrating that medical-grade accuracy is achievable with low-cost hardware plus intelligent software. Hardware cost breakdown: Raspberry Pi (€40), webcam (€10), pulse oximeter (€8), IR sensor (€5), display (€40), enclosure and power (€25). Total: €128. At production scale (500+ units), cost reduces to €100-120, representing 75-95% reduction versus commercial monitors (€500-5,000).

Conservative deployment impact estimates:

Year	Clinics	Patients/Year	Early Interventions	Lives Saved
2027	2	1,200	48	**5**
2029	30	18,000	720	**72**
2031	**200**	**120,000**	**4,800**	**480**

Calculation basis: 600 unique patients per clinic per year, 5% with abnormal vitals, 80% detection rate, 10% mortality without intervention (conservative, based on sepsis 40-50% mortality, pneumonia 5-10%).

Partnership strategy: NGOs for deployment (Partners in Health, Doctors Without Borders, AMREF). Funding sources (Gates Foundation, WHO Digital Health Network, EU Horizon Europe). Technology partners (Philips, Roche, local manufacturers in India). Regulatory pathway begins with research-only deployment, scaling through India's CDSCO (6-12 months), Kenya's approval (6 months), and WHO Prequalification (18-24 months) for global procurement.

5.3 Purpose and Vision

For two years, I built enterprise data pipelines that ensured financial data integrity. I want to apply these same data quality and validation principles to save lives. Not for those who already have access, but for the 3 billion people in rural India, Sub-Saharan Africa, and Southeast Asia who lack basic healthcare diagnostics. For the child with pneumonia whose respiratory rate goes unmeasured. For the elderly

patient with sepsis whose heart rate is not monitored. For every person who dies from a treatable condition simply because vital signs are not being tracked.

This thesis establishes the technical foundation proving that medical-grade accuracy is achievable at one-tenth the cost through intelligent software. Those 480 lives saved annually by 2031 are real people. Real children. Real parents. Real families. That is why MediSense matters as both a technical innovation and a social entrepreneurship initiative. That is the purpose driving this work.

This research bridges industrial quality control and healthcare AI to serve populations who need it most. Through rigorous technical validation and clear pathway to deployment, MediSense demonstrates that affordable, accurate vital signs monitoring for low-resource settings is not just possible: it is achievable.